

Prepared By:

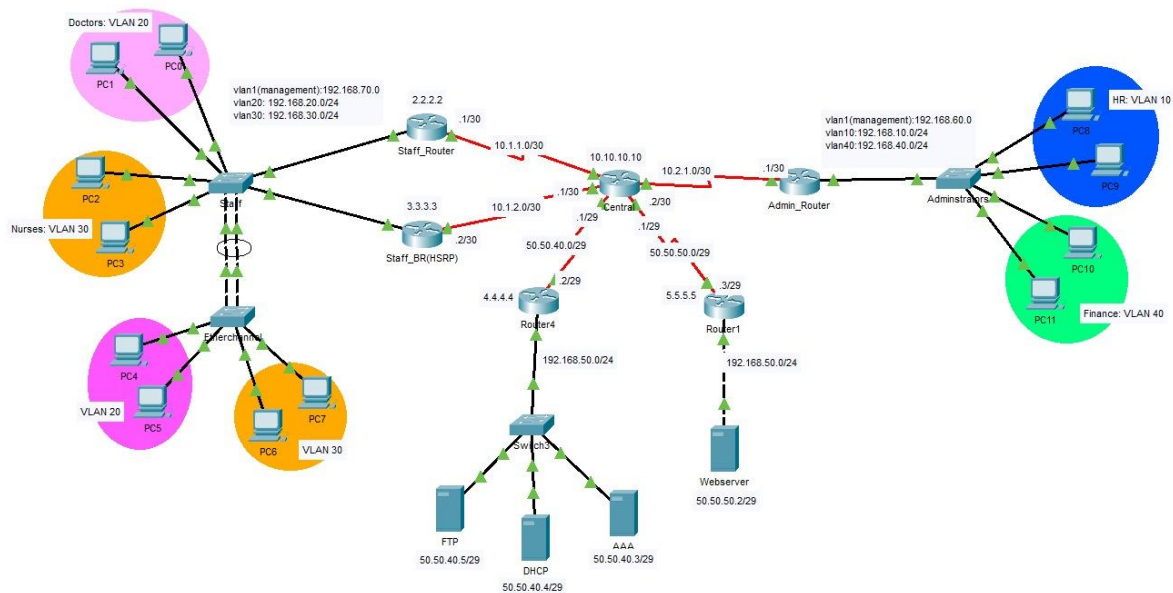
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Team Name:

Asjed

Hospital Network



Project Overview

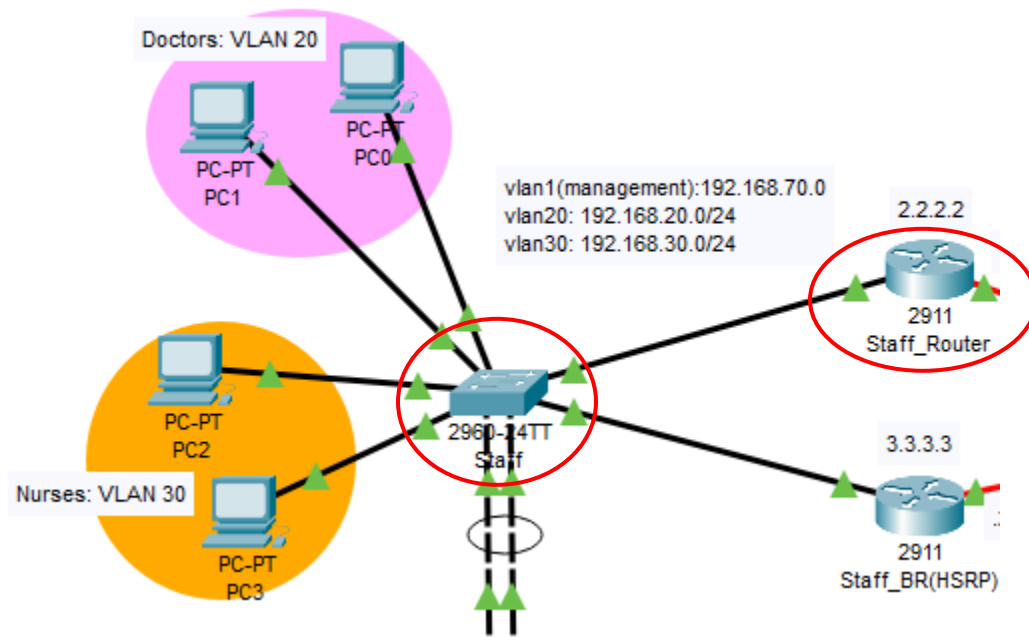
This project details the comprehensive design and configuration of a hospital network, ensuring secure, redundant, and efficient communication between various departments (Doctors, Nurses, HR, Finance, and central services).

Key technologies implemented include:

- **VLANs and Inter-VLAN Routing (ROAS):** To segment traffic for security and performance.
- **OSPF Dynamic Routing:** To enable efficient communication and route discovery across the core network segments.
- **HSRP (Hot Standby Router Protocol):** To provide redundancy and failover protection for the Staff Network gateways.
- **EtherChannel (LACP):** To improve bandwidth and redundancy between core network switches.
- **Static NAT:** For external access to the Web Server and internal servers.
- **AAA (Authentication, Authorization, and Accounting):** Using a RADIUS server for centralized device login authentication.
- **DHCP Relay:** To centralize IP address management from a single DHCP server.
- **FTP Server:** To manage and back up configuration files.
- **SSH:** Secure configuration for remote access on network devices.

Intervlan routing using ROAS

The Staff Network is designed to separate traffic between **Doctors** and **Nurses** for security and performance reasons while still allowing necessary communication between them. This is achieved by creating **two VLANs** on the same switch and enabling **inter-VLAN routing** using a router configured for **ROAS (Router-on-a-Stick)**.



ID	Name	Subnet	Gateway
VLAN 20	Doctors	192.168.20.0/24	192.168.20.1
VLAN 30	Nurses	192.168.30.0/24	192.168.30.1

Staff Switch:

```
ASJED_Staff (config)# vlan 20
ASJED_Staff (config-vlan)# name Doctors
ASJED_Staff (config-vlan)# exit
ASJED_Staff (config)# vlan 30
ASJED_Staff (config-vlan)# name Nurses
ASJED_Staff (config-vlan)# exit
ASJED_Staff (config-vlan)#
ASJED_Staff (config)# interface range fastEthernet 0/3-4
```

```

ASJED_Staff (config-if-range)# switchport mode access
ASJED_Staff (config-if-range)# switchport access vlan 20
ASJED_Staff (config-if-range)# no shutdown
ASJED_Staff (config-if-range)# exit
ASJED_Staff (config)# interface range fastEthernet 0/5-6
ASJED_Staff (config-if-range)# switchport mode access
ASJED_Staff (config-if-range)# switchport access vlan 30
ASJED_Staff (config-if-range)# no shutdown
ASJED_Staff (config-if-range)# exit
ASJED_Staff (config-if)# interface FastEthernet 0/1
ASJED_Staff (config-if)# switchport mode trunk
ASJED_Staff (config-if)# no shutdown
ASJED_Staff (config-if)# exit
ASJED_Staff (config)#exit
ASJED_Staff#show vlan brief

```

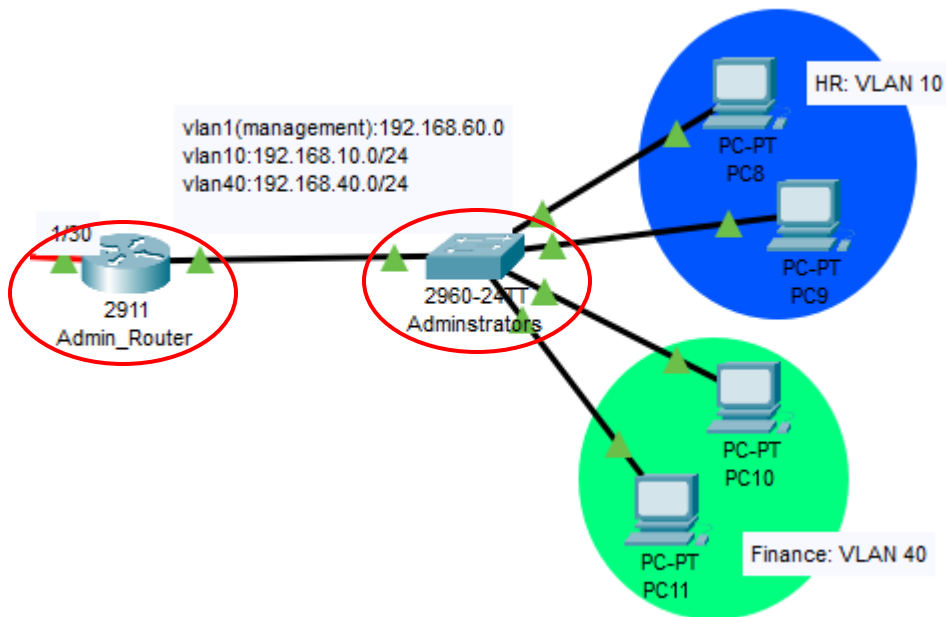
VLAN	Name	Status	Ports
1	default	active	Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
20	Doctors	active	Fa0/3, Fa0/4
30	Nurses	active	Fa0/5, Fa0/6
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Staff Router:

```
ASJED_Staff_Router (config)# interface GigabitEthernet 0/0.20
ASJED_Staff_Router (config-subif)# encapsulation dot1Q 20
ASJED_Staff_Router (config-subif)# ip address 192.168.20.1 255.255.255.0
ASJED_Staff_Router (config-subif)# no shutdown
ASJED_Staff_Router (config-subif)# exit
ASJED_Staff_Router (config)# interface GigabitEthernet 0/0.30
ASJED_Staff_Router (config-subif)# encapsulation dot1Q 30
ASJED_Staff_Router (config-subif)# ip address 192.168.30.1 255.255.255.0
ASJED_Staff_Router (config-subif)# no shutdown
ASJED_Staff_Router (config-subif)# exit
ASJED_Staff_Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	manual	up	up
GigabitEthernet0/0.20	192.168.20.1	YES	manual	up	up
GigabitEthernet0/0.30	192.168.30.1	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Serial0/0/0	10.1.2.1	YES	manual	up	up
Serial0/0/1	10.1.1.1	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

The Administration Network is divided into two VLANs to separate traffic between **Finance Department** and **HR Department**. This segmentation improves **security, performance, and management** while still allowing necessary communication through **inter-VLAN routing** on the router using **ROAS (Router-on-a-Stick)**.



ID	Name	Subnet	Gateway
VLAN 10	HR	192.168.10.0/24	192.168.10.1
VLAN 40	Finance	192.168.40.0/24	192.168.40.1

Admin Switch:

```
ASJED_Administrators (config)#vlan 10
ASJED_Administrators (config-vlan)#name HR
ASJED_Administrators (config-vlan)#exit
ASJED_Administrators (config)#vlan 40
ASJED_Administrators (config-vlan)#name Finance
ASJED_Administrators (config-vlan)#exit
ASJED_Administrators (config)#interface range FastEthernet 0/2-3
ASJED_Administrators (config-if-range)#switchport mode access
ASJED_Administrators (config-if-range)#switchport access vlan 10
ASJED_Administrators (config-if-range)#no shutdown
ASJED_Administrators (config-if-range)#exit
```

```

ASJED_Admistrators (config)#interface range FastEthernet 0/4-5
ASJED_Admistrators (config-if-range)#switchport mode access
ASJED_Admistrators (config-if-range)#switchport access vlan 40
ASJED_Admistrators (config-if-range)#exit
ASJED_Admistrators (config)#interface FastEthernet 0/5
ASJED_Admistrators (config-if)#switchport mode access
ASJED_Admistrators (config-if)#switchport access vlan 40
ASJED_Admistrators (config-if)#exit
ASJED_Admistrators (config)#interface FastEthernet 0/1
ASJED_Admistrators (config-if)#switchport mode trunk
ASJED_Admistrators (config-if)#no shutdown
ASJED_Admistrators (config-if)#exit
ASJED_Admistrators (config)#exit
ASJED_Admistrators#show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	HR	active	Fa0/2, Fa0/3
40	Finance	active	Fa0/4, Fa0/5
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

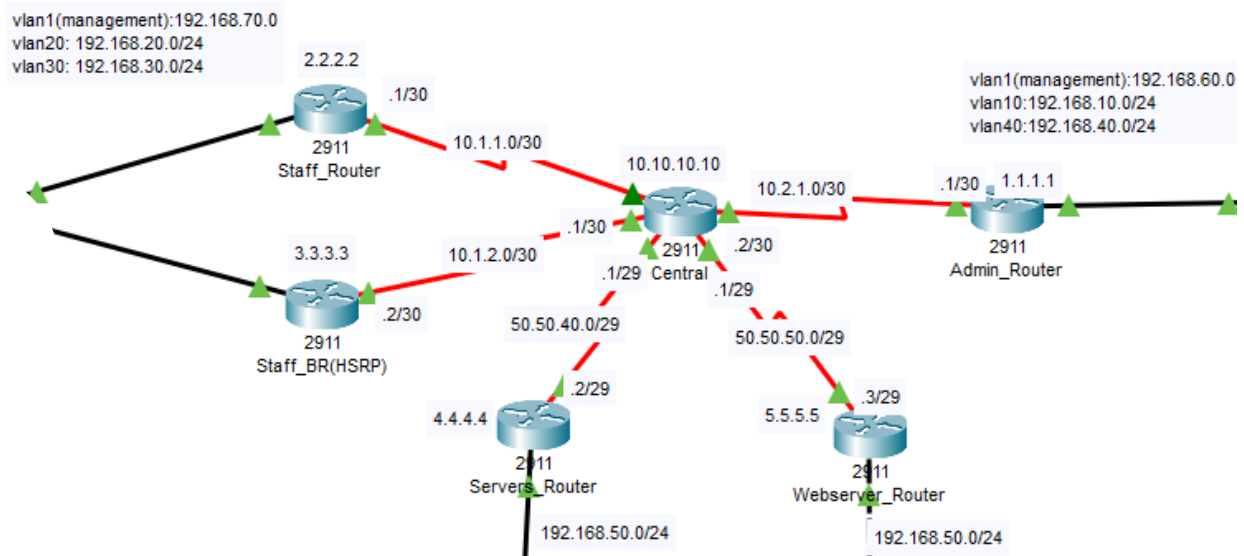
Admin Router:

```
ASJED_Admin_Router (config)#interface GigabitEthernet 0/0.10
ASJED_Admin_Router (config-subif)#encapsulation dot1Q 10
ASJED_Admin_Router (config-subif)#ip address 192.168.10.1 255.255.255.0
ASJED_Admin_Router (config-subif)#no shutdown
ASJED_Admin_Router (config-subif)#exit
ASJED_Admin_Router (config)#interface GigabitEthernet 0/0.40
ASJED_Admin_Router (config-subif)#encapsulation dot1Q 40
ASJED_Admin_Router (config-subif)#ip address 192.168.40.1 255.255.255.0
ASJED_Admin_Router (config-subif)#no shutdown
ASJED_Admin_Router (config-subif)#exit
ASJED_Admin_Router (config)#exit
ASJED_Admin_Router#
ASJED_Admin_Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	manual	up	up
GigabitEthernet0/0.10	192.168.10.1	YES	manual	up	up
GigabitEthernet0/0.40	192.168.40.1	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Serial0/0/0	10.2.1.1	YES	manual	up	up
Serial0/0/1	10.2.2.1	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

OSPF routing Protocol

To enable communication between different network segments (Staff Network, Administration Network, and other hospital departments), the routers are configured to use the **OSPF dynamic routing protocol**. OSPF ensures efficient route discovery, loop-free routing, and automatic network updates **without the need for manual static routes**.



Admin Router:

```
ASJED_Admin_Router (config)#router ospf 1
ASJED_Admin_Router (config-router)#router-id 1.1.1.1
ASJED_Admin_Router (config-router)#network 10.2.1.0 0.0.0.3 area 0
ASJED_Admin_Router (config-router)#network 192.168.10.0 0.0.0.255 area 0
ASJED_Admin_Router (config-router)#network 192.168.40.0 0.0.0.255 area 0
ASJED_Admin_Router (config-router)#network 192.168.60.0 0.0.0.255 area 0
```

```
ASJED_Admin_Router #clear ip ospf process
```

```
Reset ALL OSPF processes? [no]: yes
```

```
ASJED_Admin_Router #copy running-config startup-config
```

```
Destination filename [startup-config]?
```

```
Building configuration...
```

```
[OK]
```

ASJED_Admin_Router #show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.10.10.10	0	FULL/ -	00:00:33	10.2.1.2	Serial0/0/0

ASJED_Admin_Router #show ip route

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O   10.1.1.0/30 [110/128] via 10.2.1.2, 00:56:19, Serial0/0/0
O   10.1.2.0/30 [110/128] via 10.2.1.2, 00:56:19, Serial0/0/0
C   10.2.1.0/30 is directly connected, Serial0/0/0
L   10.2.1.1/32 is directly connected, Serial0/0/0
50.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O   50.50.40.0/29 [110/128] via 10.2.1.2, 00:56:19, Serial0/0/0
S   50.50.40.3/32 [1/0] via 10.2.1.2
O   50.50.50.0/29 [110/128] via 10.2.1.2, 00:56:19, Serial0/0/0
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L   192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
O   192.168.20.0/24 [110/129] via 10.2.1.2, 00:55:49, Serial0/0/0
O   192.168.30.0/24 [110/129] via 10.2.1.2, 00:55:49, Serial0/0/0
192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.40.0/24 is directly connected, GigabitEthernet0/0.40
L   192.168.40.1/32 is directly connected, GigabitEthernet0/0.40
192.168.60.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.60.0/24 is directly connected, GigabitEthernet0/0.1
L   192.168.60.1/32 is directly connected, GigabitEthernet0/0.1
O   192.168.70.0/24 [110/129] via 10.2.1.2, 00:55:59, Serial0/0/0
```

Staff Router:

ASJED_Staff_Router (config)#router ospf 1

ASJED_Staff_Router (config-router)#router-id 2.2.2.2

ASJED_Staff_Router (config-router)#network 10.1.1.0 0.0.0.3 area 0

ASJED_Staff_Router (config-router)#network 192.168.20.0 0.0.0.255 area 0

ASJED_Staff_Router (config-router)#network 192.168.30.0 0.0.0.255 area 0

ASJED_Staff_Router (config-router)#network 192.168.70.0 0.0.0.255 area 0

ASJED_Staff_Router#clear ip ospf process

Reset ALL OSPF processes? [no]: yes

ASJED_Staff_Router#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

[OK]

ASJED_Staff_Router#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.10.10.10	0	FULL/ -	00:00:31	10.1.1.2	Serial0/0/1
3.3.3.3	1	FULL/DR	00:00:30	192.168.70.254	GigabitEthernet0/0.1
3.3.3.3	1	FULL/DR	00:00:34	192.168.20.254	GigabitEthernet0/0.20
3.3.3.3	1	FULL/DR	00:00:36	192.168.30.254	GigabitEthernet0/0.30

ASJED_Staff_Router#show ip route

```

    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.1.1.0/30 is directly connected, Serial0/0/1
L       10.1.1.1/32 is directly connected, Serial0/0/1
O       10.1.2.0/30 [110/65] via 192.168.70.254, 01:01:43, GigabitEthernet0/0.1
          [110/65] via 192.168.20.254, 01:01:43, GigabitEthernet0/0.20
          [110/65] via 192.168.30.254, 01:01:43, GigabitEthernet0/0.30
O       10.2.1.0/30 [110/128] via 10.1.1.2, 01:01:53, Serial0/0/1
50.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O       50.50.40.0/29 [110/128] via 10.1.1.2, 01:01:53, Serial0/0/1
S       50.50.40.3/32 [1/0] via 10.1.1.2
O       50.50.50.0/29 [110/128] via 10.1.1.2, 01:01:53, Serial0/0/1
O       192.168.10.0/24 [110/129] via 10.1.1.2, 01:01:53, Serial0/0/1
192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.20.253/32 is directly connected, GigabitEthernet0/0.20
192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L       192.168.30.253/32 is directly connected, GigabitEthernet0/0.30
O       192.168.40.0/24 [110/129] via 10.1.1.2, 01:01:53, Serial0/0/1
O       192.168.60.0/24 [110/129] via 10.1.1.2, 01:01:53, Serial0/0/1
192.168.70.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.70.0/24 is directly connected, GigabitEthernet0/0.1
L       192.168.70.253/32 is directly connected, GigabitEthernet0/0.1

```

Staff HSRB:

ASJED_HSRP (config)#router ospf 1

ASJED_HSRP (config-router)#router-id 3.3.3.3

ASJED_HSRP (config-router)#network 10.1.2.0 0.0.0.3 area 0

ASJED_HSRP (config-router)#network 192.168.20.0 0.0.0.255 area 0

ASJED_HSRP Router (config-router)#network 192.168.30.0 0.0.0.255 area 0

ASJED_HSRP (config-router)#network 192.168.70.0 0.0.0.255 area 0

ASJED_HSRP#clear ip ospf process

Reset ALL OSPF processes? [no]: yes

ASJED_HSRP#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

[OK]

ASJED_HSRP#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.10.10.10	0	FULL/ -	00:00:38	10.1.2.1	Serial0/0/0
2.2.2.2	1	FULL/BDR	00:00:32	192.168.70.253	GigabitEthernet0/0.1
2.2.2.2	1	FULL/BDR	00:00:32	192.168.20.253	GigabitEthernet0/0.20
2.2.2.2	1	FULL/BDR	00:00:32	192.168.30.253	GigabitEthernet0/0.30

ASJED_HSRP# show ip route

```
10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O    10.1.1.0/30 [110/65] via 192.168.70.253, 01:11:36, GigabitEthernet0/0.1
      [110/65] via 192.168.20.253, 01:11:36, GigabitEthernet0/0.20
      [110/65] via 192.168.30.253, 01:11:36, GigabitEthernet0/0.30
C    10.1.2.0/30 is directly connected, Serial0/0/0
L    10.1.2.2/32 is directly connected, Serial0/0/0
O    10.2.1.0/30 [110/128] via 10.1.2.1, 01:11:51, Serial0/0/0
50.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O    50.50.40.0/29 [110/128] via 10.1.2.1, 01:11:51, Serial0/0/0
S    50.50.40.3/32 [1/0] via 10.1.2.1
O    50.50.50.0/29 [110/128] via 10.1.2.1, 01:11:51, Serial0/0/0
O    192.168.10.0/24 [110/129] via 10.1.2.1, 01:11:51, Serial0/0/0
192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L    192.168.20.254/32 is directly connected, GigabitEthernet0/0.20
192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L    192.168.30.254/32 is directly connected, GigabitEthernet0/0.30
O    192.168.40.0/24 [110/129] via 10.1.2.1, 01:11:51, Serial0/0/0
O    192.168.60.0/24 [110/129] via 10.1.2.1, 01:11:51, Serial0/0/0
192.168.70.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.70.0/24 is directly connected, GigabitEthernet0/0.1
L    192.168.70.254/32 is directly connected, GigabitEthernet0/0.1
```

Central Router:

ASJED_Central (config)#router ospf 1

ASJED_Central (config-router)#router-id 10.10.10.10

ASJED_Central (config-router)#network 10.1.1.0 0.0.0.3 area 0

ASJED_Central (config-router)#network 10.2.1.0 0.0.0.3 area 0

ASJED_Central (config-router)#network 10.1.2.0 0.0.0.3 area 0

ASJED_Central (config-router)#network 50.50.40.0 0.0.0.7 area 0

ASJED_Central (config-router)#network 50.50.50.0 0.0.0.7 area 0

Router#clear ip ospf process

Reset ALL OSPF processes? [no]: yes

ASJED_Central #copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

[OK]

ASJED_Central #show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	0	FULL/ -	00:00:34	10.2.1.1	Serial0/0/0
2.2.2.2	0	FULL/ -	00:00:39	10.1.1.1	Serial0/0/1
4.4.4.4	0	FULL/ -	00:00:39	50.50.40.2	Serial0/1/0
5.5.5.5	0	FULL/ -	00:00:34	50.50.50.3	Serial0/1/1
3.3.3.3	0	FULL/ -	00:00:39	10.1.2.2	Serial0/2/0

ASJED_Central #show ip route

```
10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C    10.1.1.0/30 is directly connected, Serial0/0/1
L    10.1.1.2/32 is directly connected, Serial0/0/1
C    10.1.2.0/30 is directly connected, Serial0/2/0
L    10.1.2.1/32 is directly connected, Serial0/2/0
C    10.2.1.0/30 is directly connected, Serial0/0/0
L    10.2.1.2/32 is directly connected, Serial0/0/0
50.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C    50.50.40.0/29 is directly connected, Serial0/1/0
L    50.50.40.1/32 is directly connected, Serial0/1/0
S    50.50.40.3/32 [1/0] via 50.50.40.2
C    50.50.50.0/29 is directly connected, Serial0/1/1
L    50.50.50.1/32 is directly connected, Serial0/1/1
O    192.168.10.0/24 [110/65] via 10.2.1.1, 01:13:17, Serial0/0/0
O    192.168.20.0/24 [110/65] via 10.1.2.2, 01:12:52, Serial0/2/0
      [110/65] via 10.1.1.1, 01:12:52, Serial0/0/1
O    192.168.30.0/24 [110/65] via 10.1.2.2, 01:12:52, Serial0/2/0
      [110/65] via 10.1.1.1, 01:12:52, Serial0/0/1
O    192.168.40.0/24 [110/65] via 10.2.1.1, 01:13:17, Serial0/0/0
O    192.168.60.0/24 [110/65] via 10.2.1.1, 01:13:17, Serial0/0/0
O    192.168.70.0/24 [110/65] via 10.1.2.2, 01:12:52, Serial0/2/0
      [110/65] via 10.1.1.1, 01:12:52, Serial0/0/1
```

Servers Router:

ASJED-Servers-RW (config)#router ospf 1

ASJED-Servers-RW (config-router)#router-id 4.4.4.4

ASJED-Servers-RW (config-router)#network 50.50.40.0 0.0.0.7 area 0

ASJED-Servers-RW#clear ip ospf process

Reset ALL OSPF processes? [no]: yes

ASJED-Servers-RW#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

[OK]

ASJED-Servers-RW#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.10.10.10	0	FULL/ -	00:00:35	50.50.40.1	Serial0/0/0

ASJED-Servers-RW#show ip route

```
Gateway of last resort is not set

10.0.0.0/30 is subnetted, 3 subnets
O   10.1.1.0/30 [110/128] via 50.50.40.1, 01:22:23, Serial0/0/0
O   10.1.2.0/30 [110/128] via 50.50.40.1, 01:22:23, Serial0/0/0
O   10.2.1.0/30 [110/128] via 50.50.40.1, 01:22:23, Serial0/0/0
50.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C   50.50.40.0/29 is directly connected, Serial0/0/0
L   50.50.40.2/32 is directly connected, Serial0/0/0
O   50.50.50.0/29 [110/128] via 50.50.40.1, 01:22:23, Serial0/0/0
O   192.168.10.0/24 [110/129] via 50.50.40.1, 01:22:13, Serial0/0/0
O   192.168.20.0/24 [110/129] via 50.50.40.1, 01:21:53, Serial0/0/0
O   192.168.30.0/24 [110/129] via 50.50.40.1, 01:21:53, Serial0/0/0
O   192.168.40.0/24 [110/129] via 50.50.40.1, 01:22:13, Serial0/0/0
192.168.50.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.50.0/24 is directly connected, GigabitEthernet0/0
L   192.168.50.1/32 is directly connected, GigabitEthernet0/0
O   192.168.60.0/24 [110/129] via 50.50.40.1, 01:22:13, Serial0/0/0
O   192.168.70.0/24 [110/129] via 50.50.40.1, 01:22:03, Serial0/0/0
```

Webserver Router:

Router(config)#router ospf 1

Router(config-router)#router-id 5.5.5.5

Router(config-router)#network 50.50.50.0 0.0.0.7 area 0

Router#clear ip ospf process

Reset ALL OSPF processes? [no]: yes

Router#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

[OK]

Router#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.10.10.10	0	FULL/ -	00:00:30	50.50.50.1	Serial0/0/1

Router #show ip route

Gateway of last resort is not set

```
10.0.0.0/30 is subnetted, 3 subnets
O    10.1.1.0/30 [110/128] via 50.50.50.1, 01:26:08, Serial0/0/1
O    10.1.2.0/30 [110/128] via 50.50.50.1, 01:26:08, Serial0/0/1
O    10.2.1.0/30 [110/128] via 50.50.50.1, 01:26:08, Serial0/0/1
50.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O    50.50.40.0/29 [110/128] via 50.50.50.1, 01:26:08, Serial0/0/1
S    50.50.40.3/32 [1/0] via 50.50.50.1
C    50.50.50.0/29 is directly connected, Serial0/0/1
L    50.50.50.3/32 is directly connected, Serial0/0/1
O    192.168.10.0/24 [110/129] via 50.50.50.1, 01:26:08, Serial0/0/1
O    192.168.20.0/24 [110/129] via 50.50.50.1, 01:25:48, Serial0/0/1
O    192.168.30.0/24 [110/129] via 50.50.50.1, 01:25:48, Serial0/0/1
O    192.168.40.0/24 [110/129] via 50.50.50.1, 01:26:08, Serial0/0/1
192.168.50.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.50.0/24 is directly connected, GigabitEthernet0/0
L    192.168.50.1/32 is directly connected, GigabitEthernet0/0
O    192.168.60.0/24 [110/129] via 50.50.50.1, 01:26:08, Serial0/0/1
O    192.168.70.0/24 [110/129] via 50.50.50.1, 01:25:58, Serial0/0/1
```

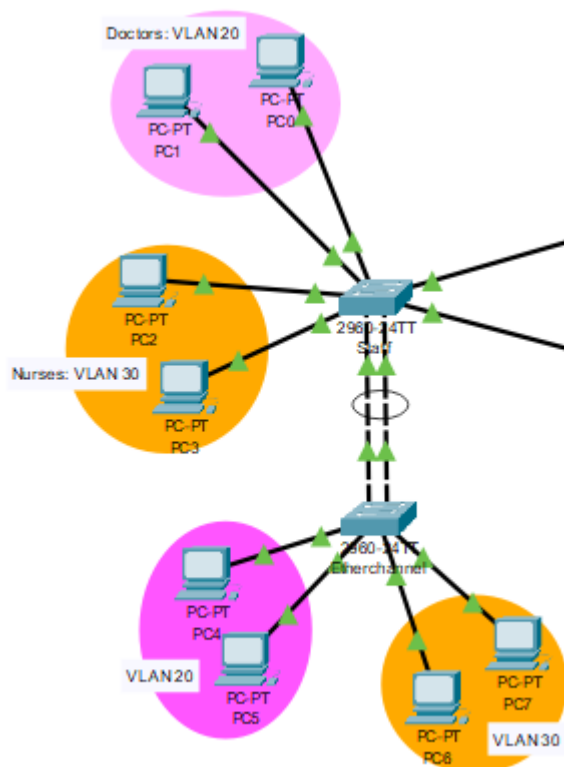
EtherChannel

To improve bandwidth, load balancing, and redundancy, the Staff Network uses **EtherChannel** to combine multiple physical links between the Staff Switch and the Router into a single **logical link**.

This ensures continuous connectivity. If one physical link fails, traffic automatically continues through the remaining active links.

EtherChannel Type:

- **LACP (Link Aggregation Control Protocol)** using **active-active** mode.



Staff Switch:

```
ASJED_Staff (config)#interface FastEthernet 0/2
ASJED_Staff (config-if)#channel-group 1 mode active
ASJED_Staff (config-if)#exit
ASJED_Staff (config)#interface FastEthernet 0/7
ASJED_Staff (config-if)#channel-group 1 mode active
```



```

ASJED_Staff (config-if)#exit
ASJED_Staff (config)#interface port-channel 1
ASJED_Staff (config-if)#switchport mode trunk
ASJED_Staff (config-if)#exit
ASJED_Staff (config)# exit
ASJED_Staff #
ASJED_Staff #show etherchannel

```

Channel-group listing:

```

Group: 1
-----
Group state = L2
Ports: 2 Maxports = 16
Port-channels: 1 Max Port-channels = 16
Protocol: LACP

```

ASJED_Staff #show etherchannel summary

```

Number of channel-groups in use: 1
Number of aggregators: 1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Pol (SU)          LACP       Fa0/2 (P) Fa0/7 (P)

```

Staff 2nd Ethernet Switch:

```

ASJED_Etherchannel (config-if)#interface range FastEthernet 0/2-3
ASJED_Etherchannel (config-if-range)#switchport mode access
ASJED_Etherchannel (config-if-range)#switchport access vlan 20
ASJED_Etherchannel (config-if-range)#exit
ASJED_Etherchannel (config)#interface range FastEthernet 0/4-5
ASJED_Etherchannel (config-if-range)#switchport mode access
ASJED_Etherchannel (config-if-range)#switchport access vlan 30
ASJED_Etherchannel (config-if-range)#exit
ASJED_Etherchannel (config)#interface FastEthernet 0/1
ASJED_Etherchannel (config-if)#channel-group 1 mode active
ASJED_Etherchannel (config-if)#exit
ASJED_Etherchannel (config)#interface FastEthernet 0/6
ASJED_Etherchannel (config-if)#channel-group 1 mode active
ASJED_Etherchannel (config-if)#exit

```

```

ASJED_Etherchannel (config)#interface port-channel 1
ASJED_Etherchannel (config-if)#switchport mode trunk
ASJED_Etherchannel (config-if)#exit
ASJED_Etherchannel (config)# exit
ASJED_Etherchannel #
ASJED_Etherchannel #show etherchannel

```

```

Channel-group listing:
-----
Group: 1
-----
Group state = L2
Ports: 2 Maxports = 16
Port-channels: 1 Max Port-channels = 16
Protocol: LACP

```

```

ASJED_Etherchannel #show etherchannel summary

```

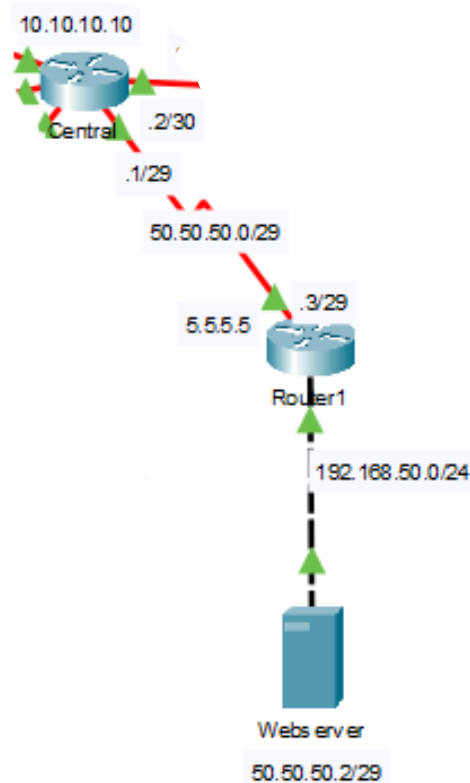
```

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Pol (SU)          LACP       Fa0/1 (P) Fa0/6 (P)

```

Web Server using Static NAT



The webserver is connected to a router (Router1) which is connected to the central router which acts as the heart of the topology that connects all the components of the Hospital network.

NetID→ **50.50.50.0/29**: is the network from which will be assigned the **inside global IP** for the webserver (which is 50.50.50.2).

NetID→ **192.168.50.0/24**: is the network from which will be assigned the **inside local IP** for the webserver. It's a private network. Therefore, it won't be from Router1's OSPF network. So, it won't be routable from outside.

Central Router:

```
interface Serial0/1/1
ip address 50.50.50.1 255.255.255.248
```

50.50.50.1 is assigned to s0/1/1 of the central router

```
router ospf 1
router-id 10.10.10.10
log-adjacency-changes
network 10.2.1.0 0.0.0.3 area 0
network 10.1.1.0 0.0.0.3 area 0
network 50.50.40.0 0.0.0.7 area 0
network 50.50.50.0 0.0.0.7 area 0
network 10.1.2.0 0.0.0.3 area 0
```

50.50.50.0 NetID is added to the OSPF network of the central router.

Webserver Router:

```
interface GigabitEthernet0/0
ip address 192.168.50.1 255.255.255.0
ip nat inside
```

192.168.50.1 is assigned to g0/0 of the webserver's router.

- ip nat inside: to specify that this interface acts as the inside side of the network.

```
interface Serial0/0/1
ip address 50.50.50.3 255.255.255.248
ip nat outside
```

50.50.50.3 IP address is assigned to s0/0/1 interface of the webserver's router.

- ip nat outside: to specify that this interface acts as the outside side of the network.

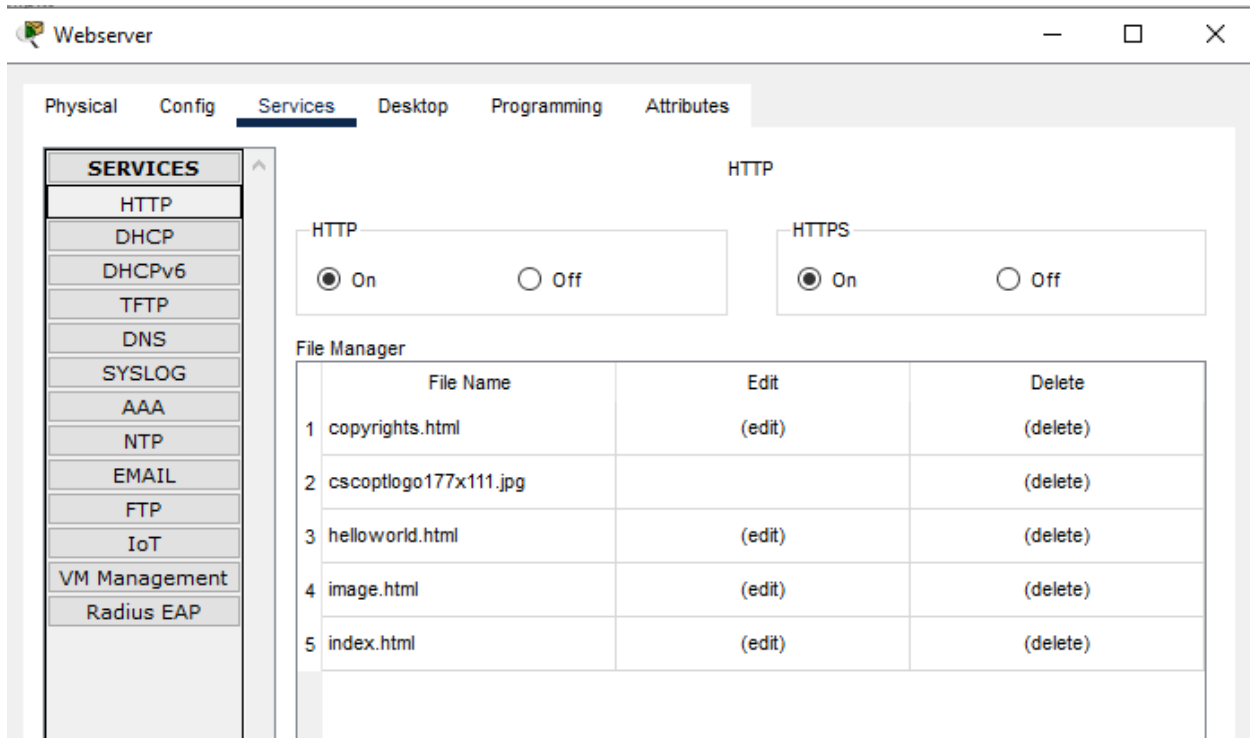
```
| ip nat inside source static 192.168.50.10 50.50.50.2
```

The command which is defining the static NAT mapping between 192.168.50.10 local IP and 50.50.50.2 global IP.

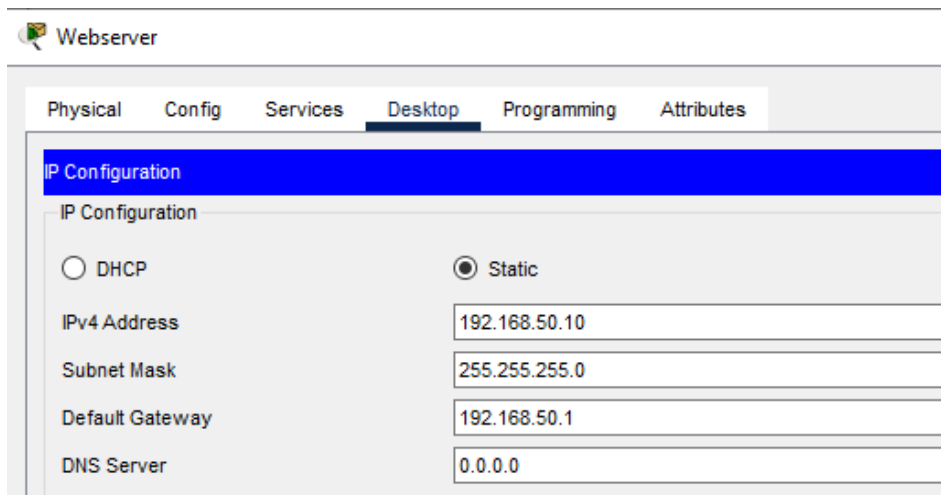
```
router ospf 1
router-id 5.5.5.5
log-adjacency-changes
network 50.50.50.0 0.0.0.7 area 0
```

50.50.50.0 NetID is added to the OSPF network of the web server router.

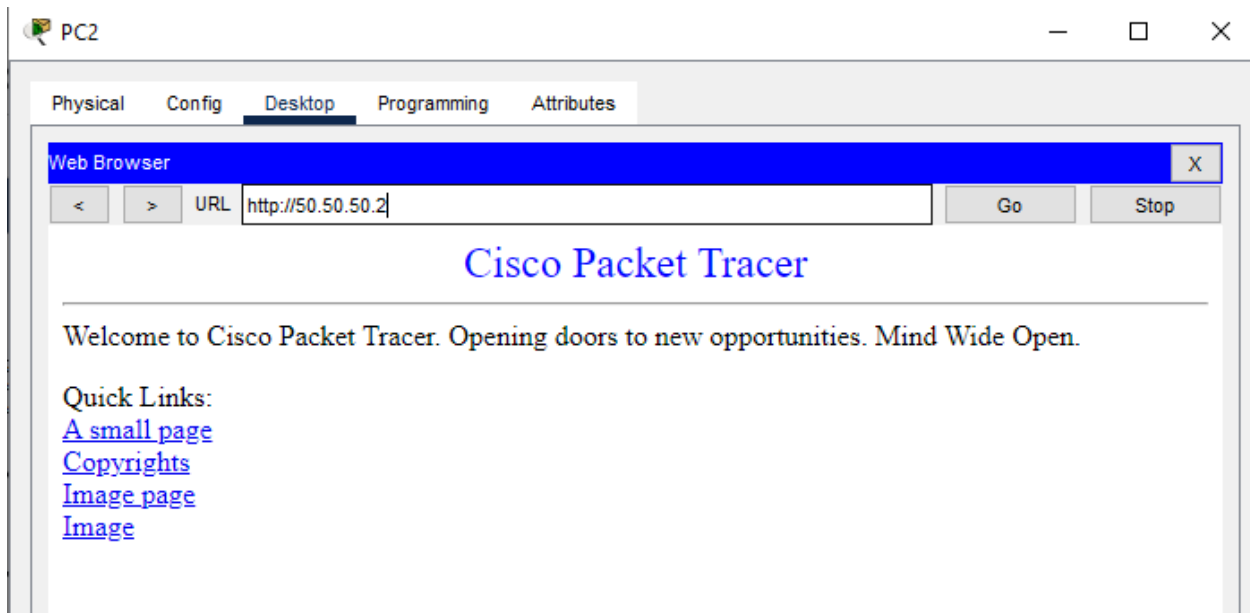
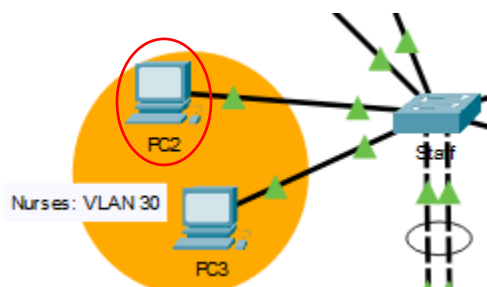
Webserver:



In the web server's services tab, we make sure the http service is enabled.

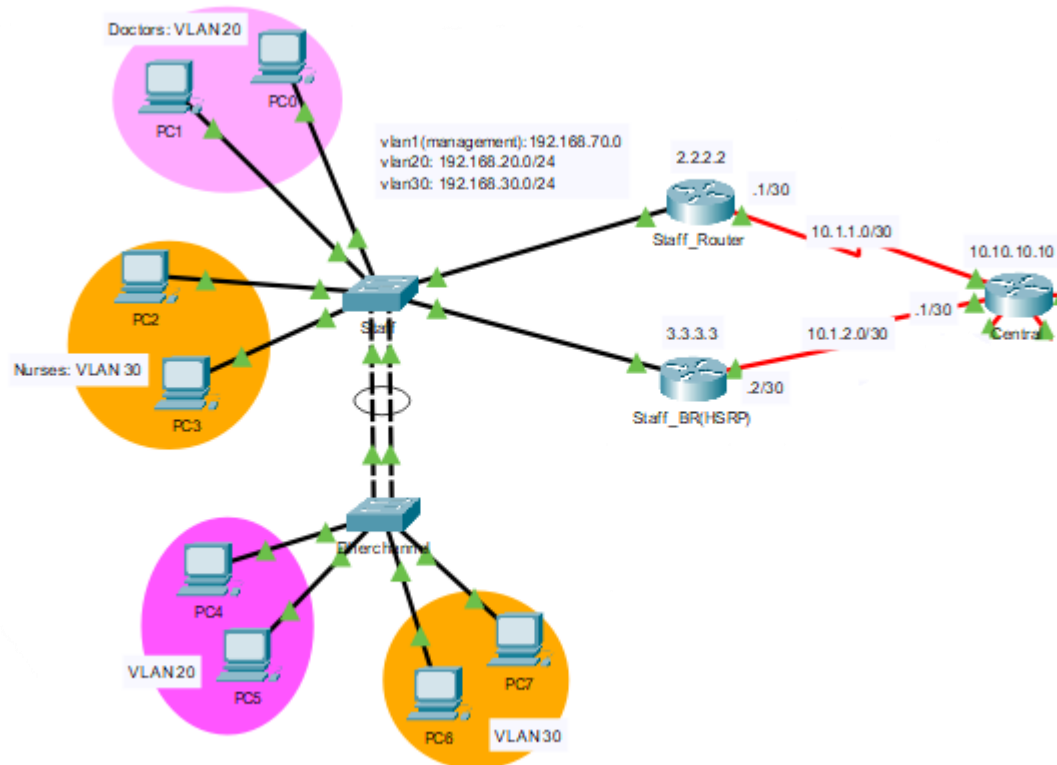


192.168.50.10 is assigned to the webserver.



PC2 connects to the server and gets the HTML page successfully.

HSRP for redundancy



The HSRP router is added for the staff network part in the topology.

Staff_router will be the primary router working.

Staff_BR(HSRP) router will be the secondary router in the standby mode if Staff_router fails.

<u>Sub-interface</u>	<u>Subnet Network ID</u>	<u>Group number</u>
G0/0.20	192.168.20.0	20
G0/0.30	192.168.30.0	30
G0/0.1	192.168.70.0	1

Staff router:

```
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.253 255.255.255.0
 ip helper-address 50.50.40.4
 standby 20 ip 192.168.20.1
 standby 20 priority 200
 standby 20 preempt
```

For g0/0.20 sub-interface, the IP assigned to it is 192.168.20.253, we made this to ensure that 192.168.20.1 is reserved for the virtual IP for the HSRP.

This sub-interface will be assigned group 20 in HSRP.

- standby 20 ip 192.168.20.1 $\Rightarrow$ this command is used to state the virtual IP.
- standby 20 priority 200 $\Rightarrow$ to make the Staff_router 0/0.20 has higher priority than the StaffBR 0/0.20 so that it works as the primary router.
- standby 20 preempt $\Rightarrow$ to force Staff_router 0/0.20 to take over if it is intact and connected properly especially after it returns from failing state.

```
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.253 255.255.255.0
 ip helper-address 50.50.40.4
 standby 30 ip 192.168.30.1
 standby 30 priority 200
 standby 30 preempt
```

For g0/0.30 sub-interface, the IP assigned to it is 192.168.30.253, we made this to ensure that 192.168.30.1 is reserved for the virtual IP for the HSRP.

This sub-interface will be assigned group 30 in HSRP.

- standby 30 ip 192.168.30.1 $\Rightarrow$ this command is used to state the virtual IP.

- standby 30 priority 200 ⇒ to make the Staff_router 0/0.30 has higher priority than the StaffBR 0/0.30 so that it works as the primary router.
- standby 30 preempt ⇒ to force Staff_router 0/0.30 to take over if it is intact and connected properly especially after it returns from failing state.

```
interface GigabitEthernet0/0.1
 encapsulation dot1Q 1 native
 ip address 192.168.70.253 255.255.255.0
 standby 1 ip 192.168.70.1
 standby 1 priority 200
 standby 1 preempt
```

G0/0.1 sub-interface is used for vlan1 interface for management purposes, for the remote connection with the switch, as well as the AAA server connection since the source IP address when the switch is communicating with the AAA server will be the vlan1 IP by default.

The IP assigned to it is 192.168.70.253, we made this to ensure that 192.168.70.1 is reserved for the virtual IP for the HSRP.

This sub-interface will be assigned group 1 in HSRP.

- standby 1 ip 192.168.70.1 ⇒ this command is used to state the virtual IP.
- standby 1 priority 200 ⇒ to make the Staff_router 0/0.1 sub-interface has higher priority than the StaffBR 0/0.1 so that it works as the primary router.
- standby 1 preempt ⇒ to force Staff_router 0/0.1 to take over if it is intact and connected properly especially after it returns from failing state.

StaffBR(HSRP):

```
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.254 255.255.255.0
 ip helper-address 50.50.40.4
 standby 20 ip 192.168.20.1
```

For g0/0.20 sub-interface for the staffBR, the IP assigned to it is 192.168.20.254, which is in the same subnet of 192.168.20.0 as StaffR.

This sub-interface will be assigned to group 20 which is the group for subnet 192.168.20.0.

- standby 20 ip 192.168.20.1 $\Rightarrow$ this command is used to state the virtual IP.

```
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.254 255.255.255.0
 ip helper-address 50.50.40.4
 standby 30 ip 192.168.30.1
```

For g0/0.30 sub-interface for the staffBR, the IP assigned to it is 192.168.30.254, which is in the same subnet of 192.168.30.0 as StaffR.

- standby 30 ip 192.168.30.1 $\Rightarrow$ this command is used to state the virtual IP.

```
interface GigabitEthernet0/0.1
 encapsulation dot1Q 1 native
 ip address 192.168.70.254 255.255.255.0
 standby 1 ip 192.168.70.1
```

The ip address assigned to g0/0.1 is 192.168.70.254 which is in the same subnet of 192.168.70.0 as staffR.

The virtual IP is 192.168.70.1.

```
ASJED_Staff_Router#sh standby
GigabitEthernet0/0.1 - Group 1
  State is Active
    5 state changes, last state change 00:00:31
  Virtual IP address is 192.168.70.1
  Active virtual MAC address is 0000.0C07.AC01
    Local virtual MAC address is 0000.0C07.AC01 (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 0.328 secs
  Preemption enabled
  Active router is local
  Standby router is 192.168.70.254, priority 100 (expires in 6 sec)
  Priority 200 (configured 200)
  Group name is hsrp-Gig-1 (default)
GigabitEthernet0/0.20 - Group 20
  State is Active
    6 state changes, last state change 00:00:31
  Virtual IP address is 192.168.20.1
  Active virtual MAC address is 0000.0C07.AC14
    Local virtual MAC address is 0000.0C07.AC14 (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 1.681 secs
  Preemption enabled
  Active router is local
  Standby router is 192.168.20.254, priority 100 (expires in 8 sec)
  Priority 200 (configured 200)
  Group name is hsrp-Gig-20 (default)
GigabitEthernet0/0.30 - Group 30
  State is Active
    4 state changes, last state change 00:00:18
  Virtual IP address is 192.168.30.1
  Active virtual MAC address is 0000.0C07.AC1E
    Local virtual MAC address is 0000.0C07.AC1E (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 2.457 secs
  Preemption enabled
  Active router is local
  Standby router is 192.168.30.254, priority 100 (expires in 7 sec)
  Priority 200 (configured 200)
  Group name is hsrp-Gig-30 (default)
ASJED_Staff_Router#
```

```

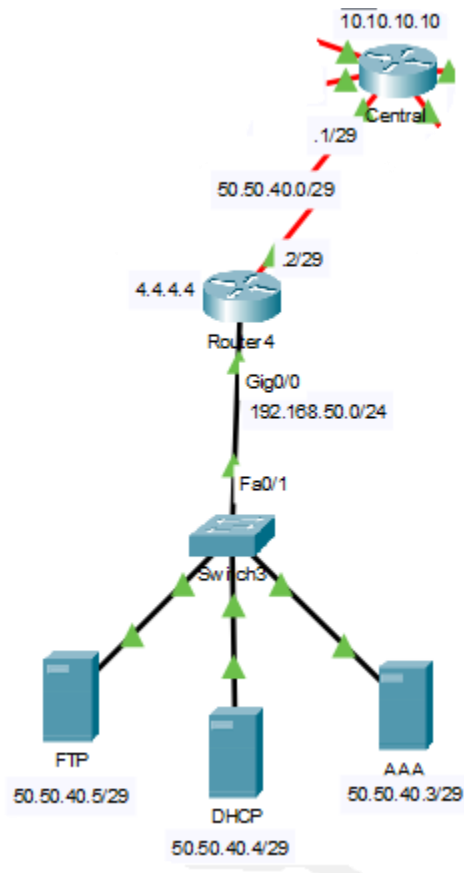
ASJED_HSRP#sh stand
GigabitEthernet0/0.1 - Group 1
  State is Standby
    11 state changes, last state change 02:29:48
  Virtual IP address is 192.168.70.1
  Active virtual MAC address is 0000.0C07.AC01
    Local virtual MAC address is 0000.0C07.AC01 (vl default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 1.375 secs
  Preemption disabled
  Active router is 192.168.70.253, priority 200 (expires in 6 sec)
    MAC address is 0000.0C07.AC01
  Standby router is local
  Priority 100 (default 100)
  Group name is hsrp-Gig-1 (default)
GigabitEthernet0/0.20 - Group 20
  State is Standby
    12 state changes, last state change 02:29:49
  Virtual IP address is 192.168.20.1
  Active virtual MAC address is 0000.0C07.AC14
    Local virtual MAC address is 0000.0C07.AC14 (vl default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 2.183 secs
  Preemption disabled
  Active router is 192.168.20.253, priority 200 (expires in 7 sec)
    MAC address is 0000.0C07.AC14
  Standby router is local
  Priority 100 (default 100)
  Group name is hsrp-Gig-20 (default)
GigabitEthernet0/0.30 - Group 30
  State is Standby
    8 state changes, last state change 02:29:48
  Virtual IP address is 192.168.30.1
  Active virtual MAC address is 0000.0C07.AC1E
    Local virtual MAC address is 0000.0C07.AC1E (vl default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 1.027 secs
  Preemption disabled
  Active router is 192.168.30.253
  Standby router is local
  Priority 100 (default 100)
  Group name is hsrp-Gig-30 (default)
ASJED_HSRP#

```

- sh standby command will show the state of the router, either active or standby.

The output of this command shows that the staff router is in active state in all the groups while the staff BR is in the standby mode.

Standalone Servers:



The servers in the 192.168.50.0 are mapped to 50.50.40.0 using static Nat.

```
ip nat inside source static 192.168.50.10 50.50.40.3
ip nat inside source static 192.168.50.20 50.50.40.4
ip nat inside source static 192.168.50.30 50.50.40.5
```

<u>Server Name</u>	<u>Inside local IP</u>	<u>Inside Global IP</u>
AAA	192.168.50.10	50.50.40.3
DHCP	192.168.50.20	50.50.40.4
FTP	192.168.50.30	50.50.40.5

(1) AAA Server

The AAA server is configured for the login authentication of the first 5 vty lines [0 to 4] for all the routers and switches existing in the topology.

Physical Config **Services** Desktop Programming Attributes

AAA

Service ☒ On ☐ Off Radius Port 1645

Network Configuration

Client Name Client IP

Secret ServerType Radius

	Client Name	Client IP	Server Type	Key	
1	ASJED_Staff_...	10.1.1.1	Radius	staffR	Add Save Remove
2	ASJED_HSRP	10.1.2.2	Radius	staffBR	
3	ASJED_Admin...	10.2.1.1	Radius	adminR	
4	ASJED_Admin...	192.168.60.2	Radius	adminsSW	
5	ASJED_Staff	192.168.70.2	Radius	staffSW	
6	ASJED_Etherc...	192.168.70.3	Radius	etherchannelSW	

User Setup

Username Password

	Username	Password	
1	admin	admin000	Add Save Remove
2	admin-1	admin123	

First, we enable AAA service in the services tab.

Secondly, we insert the usernames and passwords which are going to be the AAA server's database.

	Client Name	Client IP	Server Type	Key
1	ASJED_Staff_Router	10.1.1.1	Radius	staffR
2	ASJED_HSRP	10.1.2.2	Radius	staffBR
3	ASJED_Admin_Router	10.2.1.1	Radius	adminR
4	ASJED_Administrators	192.168.60.2	Radius	adminsSW
5	ASJED_Staff	192.168.70.2	Radius	staffSW
6	ASJED_Etherchannel	192.168.70.3	Radius	etherchannelSW
7	ASJED_Central	50.50.40.1	Radius	centralR
8	ASJED-Servers-RW	50.50.40.2	Radius	serversR
9	ASJED_WebServer-RW	50.50.50.3	Radius	webserverR

Thirdly, the network configuration for each device using the AAA server for authentication.

→ The hostname for each device is used as the client name.

→ The client IP is the source IP which the device is going to use when communicating with the AAA server.

For the router, it's going to be the IP of the interface the packet is generated from.

For the switch, VLAN 1 IP address is going to be the source IP by default.

→ The Key associated for each device.

Configuration for each device:

```
username admin secret 5 $1$mERr$7o2V98nA807p9bNxyhVTU1
username admin-1 secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/
username admin-2 secret 5 $1$mERr$UnWLGL.W18CyuGJY4mwQg/
```

The usernames and passwords for **the local database** of each device.

<u>Username</u>	<u>Password</u>
admin	admin000
admin-1	admin123
admin-2	admin456

```
aaa new-model
!
aaa authentication login auth group radius local
```

- aaa new-model: used for enabling aaa service in either router or switch
- aaa authentication login auth group radius local: this command configures the device to use the radius server and the local device as a failover backup if the radius server is unreachable.

```
enable secret 5 $1$mERr$3KIHqnoNawQFvPowLGNCN/
```

- enable secret command sets the enable secret to be able to connect via telnet or SSH.

The secret used for each device is the same as the Key configured in the AAA server for ease.

```
line vty 0 4
 login authentication auth
```

The login for the vty 0 4 is configured to use the auth authentication list.

Central Router:

```
ip route 50.50.40.3 255.255.255.255 50.50.40.2
```

This static route ensures that the communication with 50.50.40.3 will be generated from 50.50.40.1 which is the client IP for the central router configured on the AAA server.

```
radius server Rad  
  address ipv4 50.50.40.3 auth-port 1645  
  key centralR
```

Defining the radius server in the central router with its IP address and key.

Admin Router:

```
ip route 50.50.40.3 255.255.255.255 10.2.1.2
```

This static route ensures that the communication with 50.50.40.3 will be generated from 10.2.1.1 which is the client IP for the administrators router configured on the AAA server.

```
radius server Rad  
  address ipv4 50.50.40.3 auth-port 1645  
  key adminR
```

Defining the radius server in the administrators router with its IP address and key.

Staff Router:

```
ip route 50.50.40.3 255.255.255.255 10.1.1.2
```

This static route ensures that the communication with 50.50.40.3 will be generated from 10.1.1.1 which is the client IP for the staff router configured on the AAA server.

```
radius server Rad  
  address ipv4 50.50.40.3 auth-port 1645  
  key staffR
```

Defining the radius server in the staff router with its IP address and key.

Staff Backup Router:

```
ip route 50.50.40.3 255.255.255.255 10.1.2.1
```

This static route ensures that the communication with 50.50.40.3 will be generated from 10.1.2.2 which is the client IP for the staff backup router configured on the AAA server.

```
radius server Rad  
  address ipv4 50.50.40.3 auth-port 1645  
  key staffBR
```

Defining the radius server in the staff backup router with its IP address and key.

Webserver router:

```
ip route 50.50.40.3 255.255.255.255 50.50.50.1
```

This static route ensures that the communication with 50.50.40.3 will be generated from 50.50.50.3 which is the client IP for the webserver router configured on the AAA server.

```
radius server Rad  
  address ipv4 50.50.40.3 auth-port 1645  
  key webserverR
```

Defining the radius server in the webserver router with its IP address and key.

Servers Router:

```
radius server Rad  
  address ipv4 50.50.40.3 auth-port 1645  
  key serversR
```

Defining the radius server in the servers router with its IP address and key.

* No static route will be defined for this server as it is directly connected to the AAA server.

Staff Switch

```
interface Vlan1
 ip address 192.168.70.2 255.255.255.0
 ip default-gateway 192.168.70.1
```

For the staff switch, 192.168.70.2 ip address is assigned.

The default gateway is 192.168.70.1 (which is the VIP in the HSRP)

```
radius-server host 50.50.40.3 auth-port 1645 key staffSW
```

Defining the radius server in the staff switch with its IP address and key.

EtherChannel Switch:

```
interface Vlan1
 ip address 192.168.70.3 255.255.255.0
 ip default-gateway 192.168.70.1
```

For the etherchannel switch, 192.168.70.3 ip address is assigned.

The default gateway is 192.168.70.1 (which is the VIP in the HSRP)

```
radius-server host 50.50.40.3 auth-port 1645 key etherchannelSW
```

Defining the radius server in the etherchannel switch with its IP address and key.

Administrators Switch:

```
interface Vlan1
 ip address 192.168.60.2 255.255.255.0
 ip default-gateway 192.168.60.1
```

For the administrators switch, 192.168.60.2 ip address is assigned.

The default gateway is 192.168.60.1.

```
radius-server host 50.50.40.3 auth-port 1645 key adminsSW
```

Defining the radius server in the administrators switch with its IP address and key.

PC0 from the staff network is accessing the administrators router via telnet:

```
C:\>telnet 10.2.1.1
Trying 10.2.1.1 ...Open

User Access Verification

Username: admin-1
Password:
ASJED_Admin_Router>
ASJED_Admin_Router>en
Password:
ASJED_Admin_Router#
ASJED_Admin_Router#
```

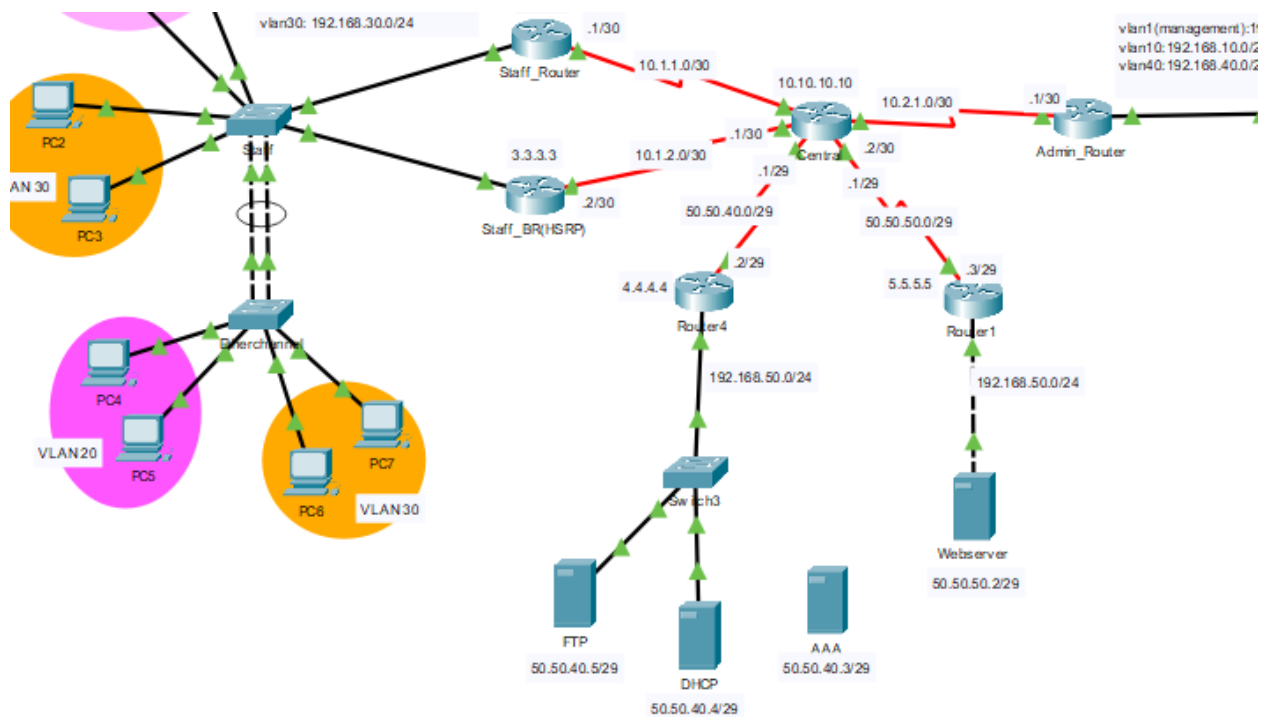
To make sure the AAA server is the one responding and not the local database, we will try to sign with admin-2 username. Admin-2 is not defined in the AAA server but defined in the local device. Hence when trying to sign in with admin-2 it should give login invalid.

```
C:\>telnet 10.2.1.1
Trying 10.2.1.1 ...Open

User Access Verification

Username: admin-2
Password:
% Login invalid

Username:
```



If the AAA server is disconnected from the network, if we try to sign in, the authentication will happen from the local DB. Hence, when we sign in with admin-2, it will take longer when connecting but eventually, it will succeed.

```
C:\>telnet 10.2.1.1
Trying 10.2.1.1 ...Open

User Access Verification

Username: admin-2
Password:
ASJED_Admin_Router>
ASJED_Admin_Router>en
Password:
ASJED Admin Router#
```

(2) FTP server

The FTP server will be used to store the configuration files of the following: the central router, the staff switch, and the administrators switch.

The screenshot shows the 'Services' tab of a network management system. The 'FTP' service is selected in the left sidebar. The main panel displays the 'FTP' configuration. The 'Service' is set to 'On'. The 'User Setup' section includes a table with four users: 'adminsSW', 'centralR', 'cisco', and 'staffSW', all with 'RWDNL' permissions. Below the table are 'Add', 'Save', and 'Remove' buttons. The 'File' section shows a list of files: 'AdminsSW-conf', 'CentralR-conf', and 'StaffSW-conf', with a 'Remove' button at the bottom right. The bottom of the window has a 'Top' button.

	Username	Password	Permission
1	adminsSW	adminsSW	RWDNL
2	centralR	centralR	RWDNL
3	cisco	cisco	RWDNL
4	staffSW	staffSW	RWDNL

	File
1	AdminsSW-conf
2	CentralR-conf
3	StaffSW-conf

In the FTP service in the services tab of the server, we will add several FTP usernames and passwords for authentication with the FTP server.

For administrators switch:

```
ip ftp username adminsSW  
ip ftp password adminsSW
```

For Staff switch:

```
ip ftp username staffSW  
ip ftp password staffSW
```

For central router:

```
ip ftp username centralR  
ip ftp password centralR
```

In the devices, an ftp username and password should also be added, they should be from the ones stated in the FTP server.

Now, the configuration file can be copied from either the switch or the router to the FTP server.

```
ASJED_Staff#copy running-config ftp:  
Address or name of remote host []? 50.50.40.5  
Destination filename [ASJED_Staff-config]? StaffSW-conf  
  
Writing running-config...  
[OK - 2051 bytes]  
  
2051 bytes copied in 0.021 secs (97000 bytes/sec)
```

```
ASJED_Central#copy running-config ftp:  
Address or name of remote host []? 50.50.40.5  
Destination filename [ASJED_Central-config]? CentralR-conf  
  
Writing running-config...  
[OK - 1940 bytes]  
  
1940 bytes copied in 0.018 secs (107000 bytes/sec)  
ASJED_Central#
```

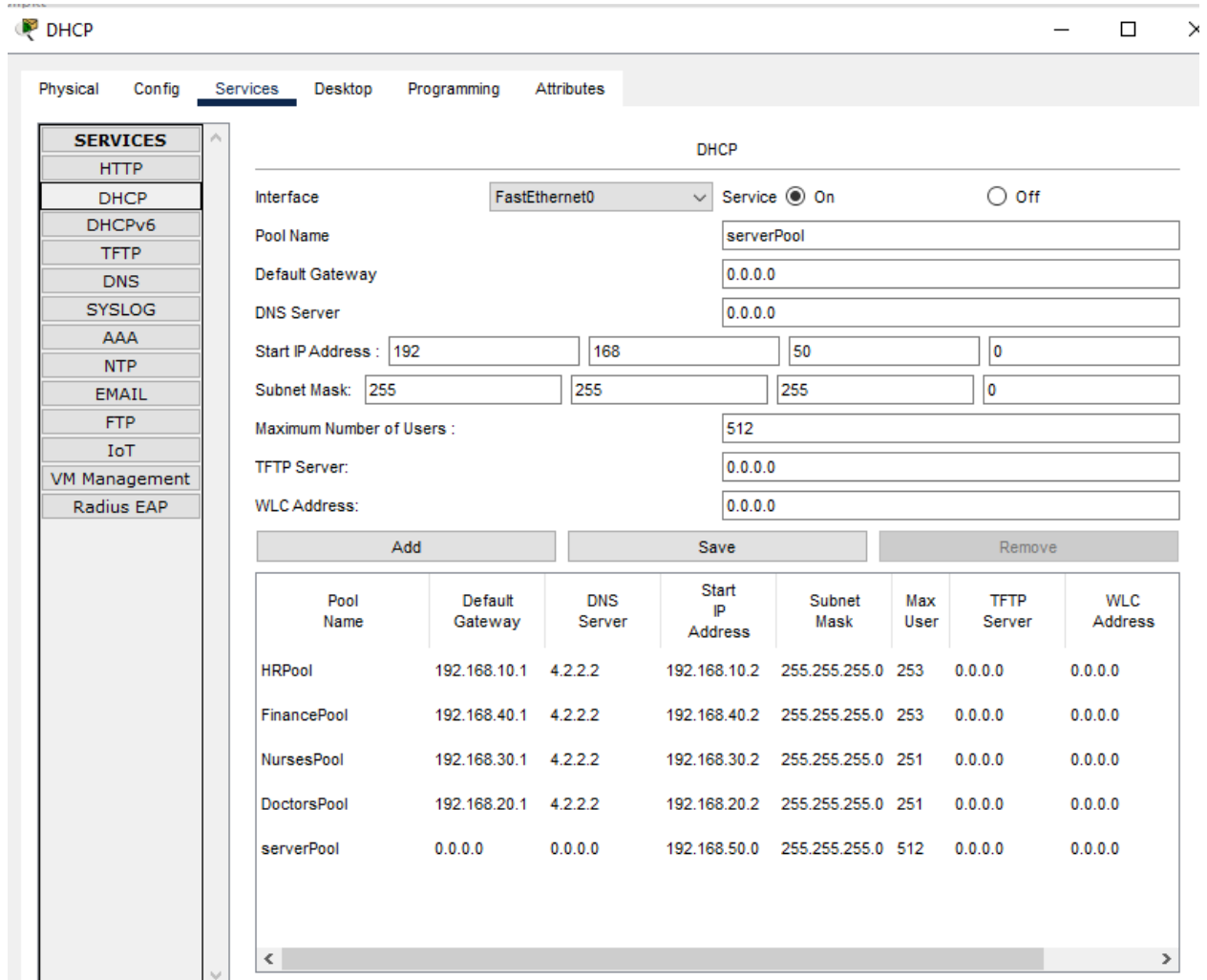
```
ASJED_Administrators#copy running-config ftp:  
Address or name of remote host []? 50.50.40.5  
Destination filename [ASJED_Administrators-config]? AdminsSW-conf  
  
Writing running-config...  
[OK - 1863 bytes]  
  
1863 bytes copied in 0.032 secs (58000 bytes/sec)  
ASJED_Administrators#
```

Here, the running-configuration is copied from the staff switch, the central router and the administrators switch to the FTP server.



The copied files can be found in the FTP server.

(3) DHCP Server



Four DHCP pools will be defined for the 4 subnets used in our topology.

→Doctors

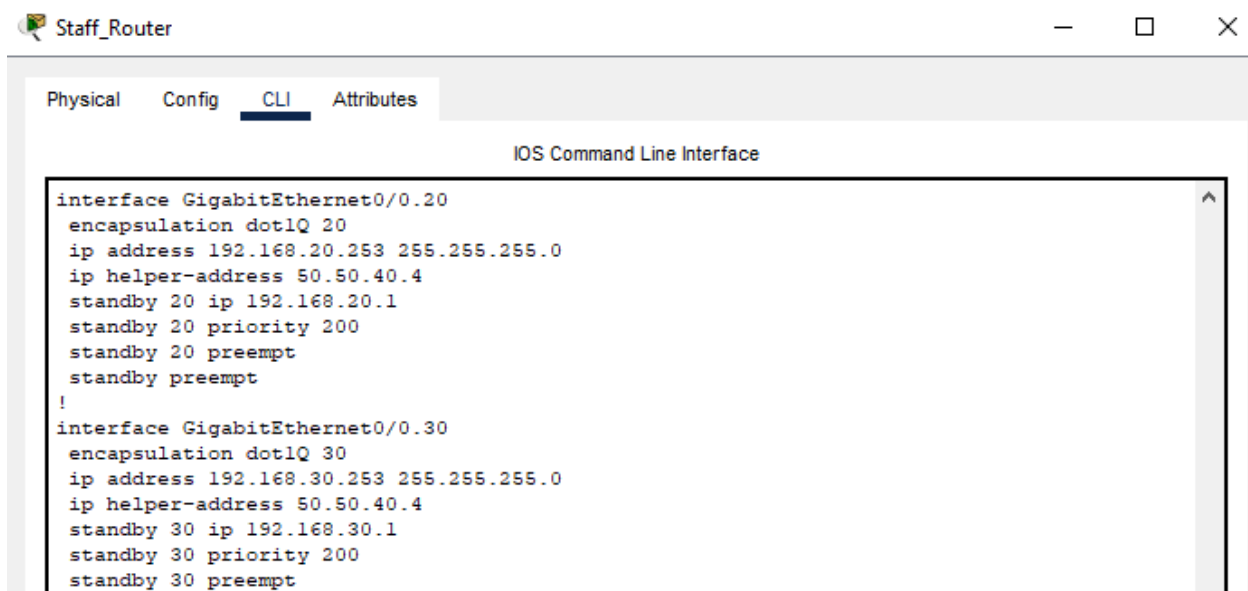
→Nurses

→Finance

→HR

Since the DHCP server is in the outside network, ip helper address command should be used for each sub-interface in the staff and the administrators router.

⇒ **Staff router:**



The screenshot shows a window titled "Staff_Router" with a tabbed interface. The "CLI" tab is selected, displaying the "IOS Command Line Interface". The configuration shown is for two interfaces, GigabitEthernet0/0.20 and GigabitEthernet0/0.30. Both interfaces are configured with encapsulation dot1Q, IP address 192.168.20.253/255.255.255.0 and 192.168.30.253/255.255.255.0 respectively, and an IP helper address of 50.50.40.4. HSRP is configured on both interfaces with group 20 for the first and group 30 for the second, both with a priority of 200 and preempt mode enabled.

```
interface GigabitEthernet0/0.20
  encapsulation dot1Q 20
  ip address 192.168.20.253 255.255.255.0
  ip helper-address 50.50.40.4
  standby 20 ip 192.168.20.1
  standby 20 priority 200
  standby 20 preempt
  standby preempt
!
interface GigabitEthernet0/0.30
  encapsulation dot1Q 30
  ip address 192.168.30.253 255.255.255.0
  ip helper-address 50.50.40.4
  standby 30 ip 192.168.30.1
  standby 30 priority 200
  standby 30 preempt
```

⇒ **Staff Backup router (HSRP):**

```
interface GigabitEthernet0/0.20
  encapsulation dot1Q 20
  ip address 192.168.20.254 255.255.255.0
  ip helper-address 50.50.40.4
  standby 20 ip 192.168.20.1
!
interface GigabitEthernet0/0.30
  encapsulation dot1Q 30
  ip address 192.168.30.254 255.255.255.0
  ip helper-address 50.50.40.4
  standby 30 ip 192.168.30.1
```

⇒ **Administrators router:**

```
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 ip helper-address 50.50.40.4
!
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 192.168.40.1 255.255.255.0
 ip helper-address 50.50.40.4
.
```

- ip helper-address 50.50.40.4: It enables the router as a DHCP relay agent to forward the DHCP broadcast traffic from the client, and convert it into a unicast message, the unicast packet's source IP address is set to the router's IP address on the client's local interface, ensuring the DHCP server (50.50.40.4) can send the response back to the correct network.

PC0 from vlan 20 on staff router:

PC0

Physical	Config	Desktop	Programming	Attributes
IP Configuration				
Interface		FastEthernet0		
IP Configuration				
☒ DHCP		☐ Static		
IPv4 Address		192.168.20.4		
Subnet Mask		255.255.255.0		
Default Gateway		192.168.20.1		
DNS Server		4.2.2.2		

PC2 from Vlan 30 on staff router:

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.168.30.4

Subnet Mask 255.255.255.0

Default Gateway 192.168.30.1

DNS Server 4.2.2.2

PC8 from Vlan 10 on administrators router:

PC8

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

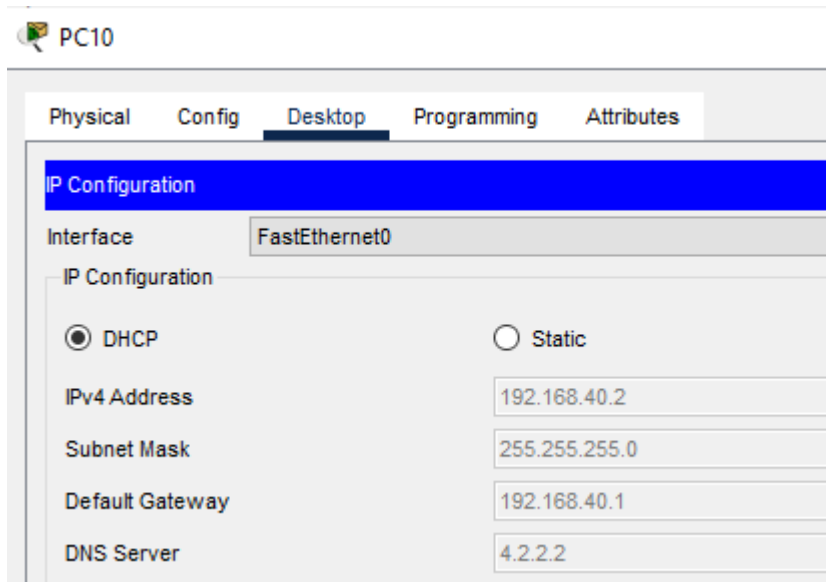
IPv4 Address 192.168.10.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 4.2.2.2

PC10 from Vlan 40 on the administrators router:



All the PCs from different Vlan on different switches succeeded to attain an ip address from the DHCP server.

SSH Configuration:

The SSH Configuration is done on the central router only.

SSH configuration conditions:

- 1) Enable secret enabled
- 2) Username & password
- 3) Encryption key generation
- 4) Enabling SSH on the vty lines.

} Already configured during the AAA setup.

Encryption key generation:

```
hostname ASJED_Central

ip domain-name ASJED-hospital.com

ASJED_Central(config)#crypto key generate rsa
% You already have RSA keys defined named ASJED_Central.ASJED-hospital.com .
% Do you really want to replace them? [yes/no]: yes
The name for the keys will be: ASJED_Central.ASJED-hospital.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

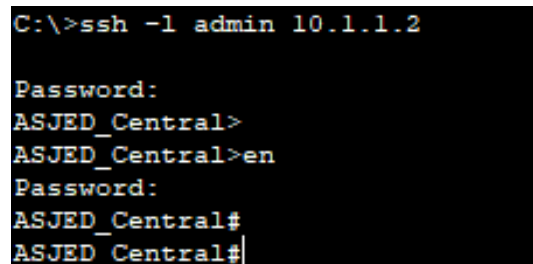
ASJED_Central(config)#
```

- crypto key generate rsa: is the command used for generating the key.

Enabling SSH on the vty lines:

```
line vty 0 4
 login authentication auth
 transport input ssh
.
```

- transport input ssh: for enabling SSH.



```
C:\>ssh -l admin 10.1.1.2

Password:
ASJED_Central>
ASJED_Central>en
Password:
ASJED_Central#
ASJED_Central#
```

When accessing central router from a PC, it works correctly.